Fakarny

-a smart note taking app for egyptian arabic-

Abstract

Automatic Speech Recognition systems have achieved remarkable accuracy in recent years; however, their performance degrades significantly when applied to regional dialects and informal code-switching.

This happens frequently here in Egypt, everyday communication relies heavily on Egyptian Arabic, which is frequently used with with English technical terms and conversational filler words. Standard ASR models trained predominantly on Modern Standard Arabic fail to accurately transcribe these nuances, making hands-free note-taking inefficient for local egyptian users.

This project proposes, an end-to-end deep learning pipeline designed to convert unstructured, rambling Egyptian spoken audio into clean, structured notes and action items. The system comprises two main stages:

1. **Dialect-Aware ASR:** Fine-tuning a pre-trained sequence-to-sequence audio model on local Egyptian speech datasets to generate accurate colloquial transcriptions.
2. **NLP Text Structuring:** Fine-tuning a regional language model to filter out conversational filler words, normalize code-switched terms, and format the core ideas into bulleted summaries and action items.

Problem Statement & Motivation

Problem Statement

While modern Automatic Speech Recognition (ASR) systems such as : OpenAI's Whisper or Google Voice, achieve high accuracy for dominant global languages and Modern Standard Arabic (MSA), they struggle with regional Arabic dialects.

In everyday communication, Egyptians speak Dialectal Egyptian Arabic, which is different from MSA in phonetics, vocabulary, and grammar. Furthermore, Egyptian speakers frequently use code-switching (mixing Arabic and English mid-sentence) alongside conversational filler phrases.

When existing speech recognition models process this informal, rambling speech, two major issues arise:

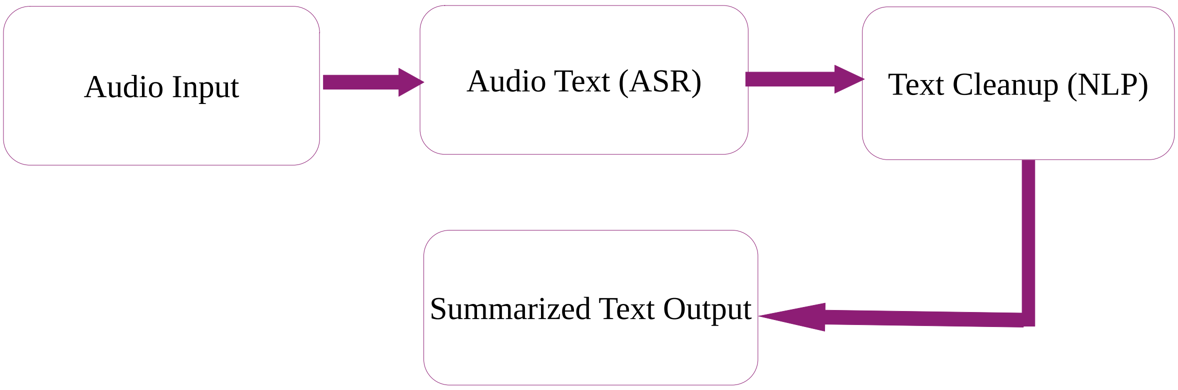
1. **Transcription Failure:** Standard models attempt to map dialectal phonemes onto formal MSA words, leading to high Word Error Rates (WER) or complete hallucination of text.
2. **Information Noise:** Even if speech is transcribed verbatim, raw voice notes remain disorganized, repetitive, and cluttered with filler words, making them impractical for quick review or task tracking.

Motivation

In Egypt, voice messaging is often preferred over manual typing for quick communication across both professional and personal contexts. However, transforming a fast-paced, 2-minute voice message into usable task lists or study notes currently requires manual listening and transcription.

Developing a dialect-aware deep learning pipeline bridges this technology gap. By tailoring ASR and NLP models specifically to egyptian dialect, this project provides a localized productivity solution: allowing users to speak naturally (rambling, slang, and code-switching included), while automatically outputting organized, actionable text summaries. Beyond the note-taking application, this project contributes to the broader effort of improving language accessibility and NLP tools for non-standard Arabic dialects.

Methodology



Datasets & Tools

We found that to implement this system efficiently without building models from scratch, we can rely on open-source resources.
For training and evaluation, we can use publicly available Egyptian Arabic speech datasets from Hugging Face, such as **MAdel121/arabic-egy-cleaned** or the **Egyptian-ASR-MGB-3** dataset, which feature real-world colloquial Egyptian speech.
For the software stack, we can use **Python** and the **Hugging Face Transformers** library to access lightweight, pre-trained models (such as Whisper-small for speech-to-text and AraT5 for text summarization).
Finally, to demonstrate the project interactively, we can build a simple web interface using **Gradio** or **Streamlit**, allowing users to record their voice and view the generated notes in real time.

These are all just expectations based on our current knowledge and limited research and may change in the future with new knowledge.